

Hipper — Technical Documentation

Team Digital Life - Hipper Monitoring

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DIGITAL
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1. Base Station

To actually get the data from the sensors to the users, a Raspberry Pi is used. In this example, a Raspberry Pi 4b is used. This device uses Bluetooth to get the data from the PAM monitors and get it visible to the client.

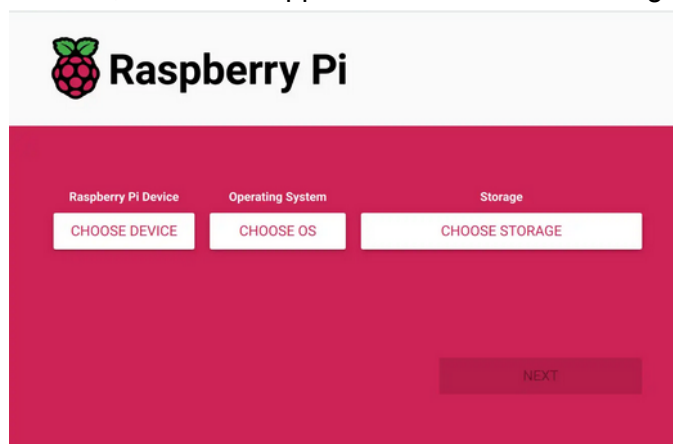
1.1 Base Station setup

To get started using the base station, you will need a few things:

- Raspberry Pi 4b (Or any other Wi-Fi and Bluetooth capable device that can run python scripts and pull a branch from git, like for example a Raspberry Pi zero 2 W). This will be the base station.
- Way to power the base station.
- SD card with at least 4 GB of storage for the base station.
- Computer with a way to connect the SD card.
- A Wi-Fi connection with the computer.

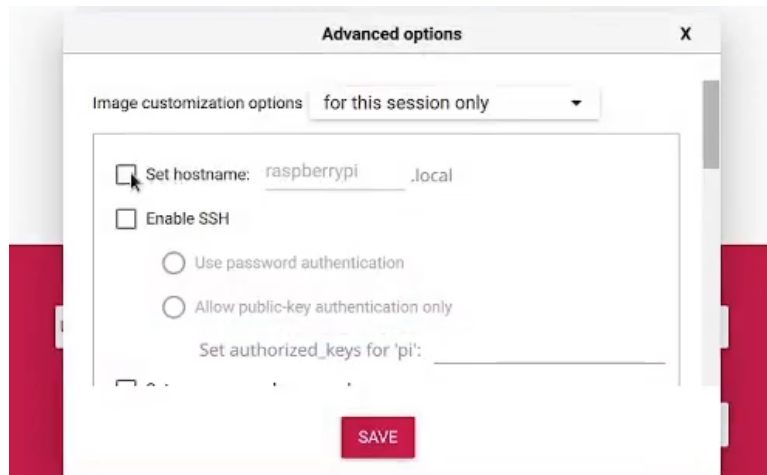
1.1.1 Setting up Raspberry Pi

To start working with the base station, we will first need to set up the Raspberry Pi. For this, take the SD card and connect it to your computer. Then go to <https://www.raspberrypi.com/software/> to install the Raspberry Pi imager. Once the imager is installed, run it. The application will look something like this:



Choose the correct Raspberry Pi device you are using, in this case, the Raspberry Pi 4b. Select an operating system from Raspberry Pi OS (other) and pick the Raspberry Pi OS Lite (64bit). Choose the ID card as the storage and press next.

Then go to advanced options:



Make sure you enable Set hostname and set the hostname as: hippy.

also enable SSH, with use of password authentication enabled.

Then create a user named hippy.

You can also set up a Wi-Fi connection here with the connection the pi should have. This can always be changed later.

Now you can flash the SD card. Once flashed, eject the card and put it into the Raspberry Pi. You can now power up the Raspberry Pi and either use an ethernet cable or be connected to the same network as the Raspberry Pi to continue with the setup.

1.1.2 Using project functions

To use the project functionalities, you must first have a Raspberry Pi that has been fully setup. This can be found in 8.2.1. Once this has been done, you can continue with the setup to get the base station fully functioning.

With a setup pi, use your computer to log in to the Raspberry Pi with ssh. This should be `ssh hippy@hippy.local`. Fill in the password of the user to login. When logged in, we need to make sure git and python is installed to get the code for this project. We can do this by running the following commands:

```
sudo apt update
```

```
sudo apt install git -y
```

```
sudo apt install -y python3-pip python3-venv
```

Once you confirm git has been installed, we can pull the git repository and start with the setup:

```
git clone
https://gitlab.fdmci.hva.nl/IoT/2024-2025-semester-2/group-project
/duuseedeewuu36.git

cd duuseedeewuu36
```

Once the repository has been pulled and we are in the correct folder, we need to make sure that the raspberry pi has the correct link to the back-end in the code. open the main code by running:

```
nano src/back-end/pam/main.py
```

At the top on line 13, place the link to where the main page of your project is located. If this is local, it will be [http://\[local_IP_address\]:5000](http://[local_IP_address]:5000). Though it would be better to have a static IP address where the page and back-end is located. Save the file and exit.

Now that this is all correct, we can run the setup file by running:

```
chmod +x pi_setup.sh

./pi_setup.sh
```

This will install all the libraries needed and create a service that will run the code on startup. Once done, you can check if the service is online by running:

```
sudo systemctl status pam.service
```

And you can see the logs of this service by running:

```
journalctl -u pam.service -f
```

Your base station is now fully set up.

1.2 Base Station functionality

Once the base station has been set up, it will function on its own once you power it up. When powered up it will automatically start the code and start trying to collect data. Note that it does require a WiFi connection to do this.

If it does have a WiFi connection, the base station will start scanning for low powered bluetooth devices that correspond with a PAM monitor. If it finds 1, it will check if the mac address of this device is in the database. If not it will skip trying to pull data from this monitor, but if it is it will check the last time minute data was pulled, and day data was pulled. If the day data has not been pulled for at least 1 day, it will attempt to pull this data and put this in the database. If the minute data has not been pulled for at least 1 hour, it will attempt to pull this data and put it in the database. If these criteria are not met it will not pull any data.

2. Setting up a local version of the Hipper Therapeutics web application

To get the web application up and running, there are 3 critical software components required: Git, VS Code and Docker. For each of these components, a general way of installing and setting up will be provided, but then tailored specifically to the Hipper Project. A dumbed down version of the installation:

- Installing VS Code
- Installing Git
- Cloning the project in VS Code through Git
- Installing Docker
- Executing the Docker script to launch the Hipper Therapeutics web application

2.1 Installing VS Code

1. Download the Installer

- Go to the official website: <https://code.visualstudio.com/>
- Click the "**Download for Windows**" button

2. Run the Installer

- Locate the downloaded **.exe** file (usually in your Downloads folder) and double-click it.

3. Follow the Installation Wizard

- Accept the license agreement.
- Choose the destination folder (default is fine).
- Check the boxes.
- Click **Next** and then **Install**.

4. Launch VS Code

- After installation, check "Launch Visual Studio Code" and click **Finish**.

2.2 Installing Git

2.2.1 Windows

1. Go to <https://git-scm.com/downloads>
2. Click "**Download for Windows**"
3. Open the `.exe` file and follow the installer:
 - Leave all of the default options checked.
 - Make sure "**Git from the command line**" is selected.
4. After installation, open **Git Bash** or **Command Prompt (CMD)** and type:

```
git --version
```

You should see something like: `git version 2.xx.x`

2.2.2 macOS

1. **Open Terminal**
 - Press `Cmd + Space`, type `Terminal`, and hit Enter.
2. **Install Xcode Command Line Tools**

In the Terminal, type:

```
xcode-select --install
```

A popup will appear:

- Click **Install** when prompted.
- Wait for the installation to finish.

3. **Verify the installation**

After it's done, check Git by typing:

```
git --version
```

You should see something like: `git version 2.xx.x`

2.3 Cloning the project in VS Code

1. **Open VS Code.**
2. **Click on the Source Control icon** on the left sidebar (or press **Ctrl + Shift + G**). 
3. **Click “Clone Repository”** button at the top (or “Clone Repository from GitHub” if using GitHub).
4. **Paste the Git repository URL**, then press **Enter**.
5. **You may be prompted to log in to one of your Git accounts**, enter your credentials when prompted.
6. **Choose a folder** on your computer where you want to save the project.

2.4 Installing Docker

2.4.1 Windows

1. **System Requirements**
 - Windows 10 or 11 (Pro, Enterprise, or Education) with **WSL2** enabled.
 - Enable **Virtualization** in BIOS.
2. **Install Docker Desktop**
 - Go to: <https://www.docker.com/products/docker-desktop/>
 - Download for Windows.
 - Run the installer and follow prompts.
3. **Post-Install**
 - Launch Docker Desktop.
 - It will install the WSL2 backend if not already installed.
 - Verify the install by putting the following in CMD or PowerShell:

```
docker --version
```

2.4.2 macOS

1. **System Requirements**
 - macOS 11+ (Big Sur or later)
 - Apple Silicon (M1/M2) or Intel
2. **Install Docker Desktop**
 - Go to: <https://www.docker.com/products/docker-desktop/>
 - Download the macOS version (choose Intel or Apple Chip).
 - Open the **.dmg** file and drag Docker to Applications.
 - Launch Docker.
3. **Verify Installation**

```
--version
```

2.5 Executing the Hipper Therapeutics Docker script

1. Open the project in VS Code

Launch VS Code, go to **File** → **Open Folder**, and select the folder where you cloned the Hipper Therapeutics project.

2. Find the `docker-compose.yml` file

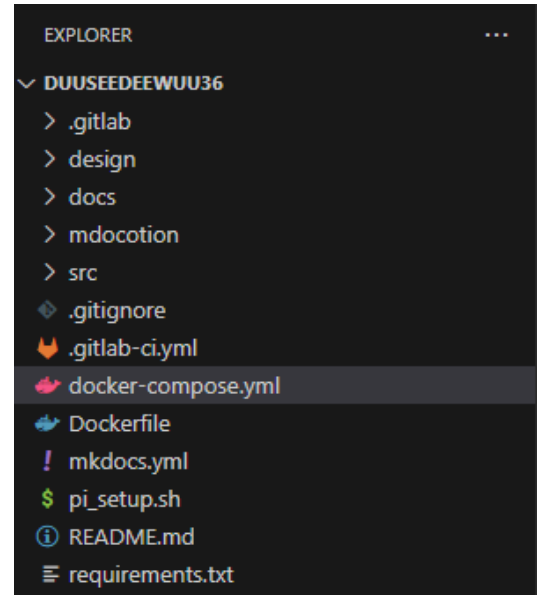
Look in the file explorer (left panel) for a file named `docker-compose.yml`. This file should be located in the root of the project.

3. Open the terminal in VS Code

Go to **Terminal(top left)** → **New Terminal** or press **Ctrl +** (Windows/Linux) or **Cmd +** (Mac). This opens a terminal window at the root of your project.

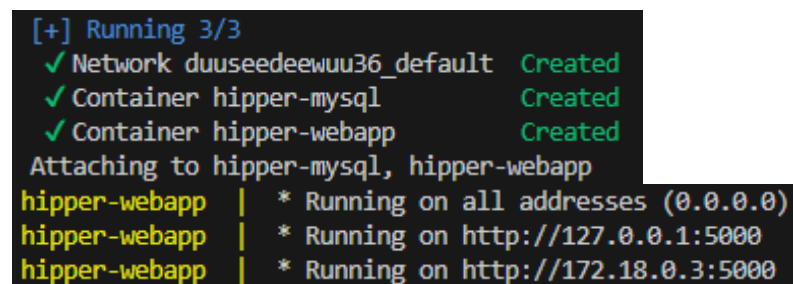
4. Run the Docker Compose script

In the terminal, type the command visible below and press Enter. This will start building the Hipper database and launch that together with the Flask web application.



```
docker-compose up
```

After executing the command you should immediately be greeted with the following:



Now simply navigate to the login page of the web application via:

- <http://localhost:5000>
- <http://127.0.0.1:5000>

Note: This URL (<http://172.18.0.3:5000>) is dynamic and will change every time you start the service.